A positive-channel wire for deep softplus posteriors

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Abstract

For the standard softplus activation, a bounded-weight shallow Boolean classifier can be transmitted through many hidden layers at isotropic Gaussian energy $L + O(\log^2(LT))$, where T is polynomial in the shallow teacher resources. This matches the leading coefficient of the nonconstant energy lower bound. The construction uses actual constant neurons and charges their injection and final subtraction. Its operator stability is polynomial even though the constant-channel weights depend on a logarithmic anchor. Combined with a Lipschitz representation cover and a Gaussian-distance posterior bound, it gives quasipolynomial sample complexity at every finite learner width in cold limits and at every fixed variance exponent $\gamma > 1$, once depth exceeds a squared logarithm of the teacher resources. The critical exponent $\gamma = 1$, arbitrary deep teachers, and the original tanh question remain outside this result.

1 Model, theorem, and proof dependencies

Let $F(a) = \log(1 + e^a)$, $X_d = \{-1, 1\}^d$, and $d \geq 1$. A learner has $L \geq 2$ matrices, $L - 1$ softplus hidden layers of common width N, augmented input $(1, x)$, no internal biases, and a linear output $z_W(x)$. The independent Gaussian entry variances in the first matrix, internal matrices, and readout are respectively

$$t^2/d, \qquad t^2/N, \qquad t^2/N^2, \qquad 0 < t \leq 1.$$

The scale may be prescribed as $t = t(d, L, N)$. For each fixed $\zeta > 0$, the theorem is simultaneous over $0 < t \leq N^{-\zeta/2}$, and therefore includes every such varying choice, in particular $t(N) = N^{(1-\gamma)/2}$ with fixed $\gamma = 1 + \zeta > 1$. For a target-labeled sample $S = ((x_i, f(x_i)))_{i=1}^n$, set

$$E_S = \{|z_W(x)| > 1/2 \ \forall x \in X_d\} \cap \{\operatorname{sign} z_W(x_i) = f(x_i) \ \forall i\}, \qquad Q_S = \mu(\cdot \mid E_S).$$

The confidence-failure mass under uniform cube inputs is

$$e(S) = 2^{-d} \#\{x : Q_S(\operatorname{sign} z_W(x) = f(x)) \leq 2/3\}.$$

Assume the target has a one-hidden-layer teacher of width C, physical weights bounded by $B \geq 1$, and signed output at least one everywhere. Choose an integer

$$M_0 = \left\lceil \max\{C, 2, d + 1, dC(d + 1)B^2\} \right\rceil, \qquad T = M_0 C B^2 \geq 1.$$

These quantities are polynomial in the teacher resources.

Theorem 1.1. *There are universal constants c_0, c_1 such that, if*

$$L - 1 \geq c_0 \log^2(T + 2),$$

then every width $N \geq C$ has the following learning guarantee. Put $R = 2 + d + C + B + L + \varepsilon^{-1} + \log(2/\delta)$, with $0 < \varepsilon, \delta < 1$. Every limiting sign posterior as $t \downarrow 0$, at a fixed finite architecture, satisfies

$$n \geq \exp[c_1 \log^2 R] \quad \Longrightarrow \quad \mathbb{P}_S\{e(S) > \varepsilon\} \leq \delta.$$

For every fixed $\zeta > 0$, the same holds for the actual finite-scale posterior, with c_1 replaced by a constant c_ζ, simultaneously for all $N \geq C$ and $0 < t \leq N^{-\zeta/2}$. Thus the common isotropic variance is at most $N^{-(1+\zeta)}$.

A limiting sign posterior means any subsequential limit on the finite set of Boolean truth tables; the architecture is fixed before taking that limit. The theorem is an upper bound for a shallow target class, and does not bound arbitrary deep teachers by their width and depth.

We use one representation lemma from the logistic companion [1]: on a finite input set of radius at most two and log-cardinality $O(d)$, globally margined networks with arbitrary zero-fixing 1-Lipschitz scalar activations, depth L, product of Frobenius norms at most $P \geq 1$, and final offset at most $2P$ have a common finite sign class with log-cardinality $\mathrm{poly}(d, L, P)$. Its proof compresses each layer by row-column importance sampling, then stores finite scalar activation tables. It is independent of the original width and requires no precision assumption on the original weights or activations. Below we provide the softplus wire, its stability, and the posterior transfer, including the small-width case.

2 A matching-energy wire

Use energy coordinates

$$A_1 = \sqrt{d/N}\, W_1, \quad A_j = W_j \ (2 \leq j < L), \quad A_L = \sqrt{N}\, v, \qquad \kappa(A) = \sum_{j=1}^{L} \|A_j\|_F^2 .$$

All coordinates have variance $\sigma^2 = t^2/N$. The normalized input is $(1, x)/\sqrt{d}$, hidden states are $h_j/\sqrt{N}$, and the normalized activation is $F(\sqrt{N}a)/\sqrt{N}$. Since F is 1-Lipschitz, a nonconstant globally absolute-margined classifier has, on some opposite-label cube edge,

$$\prod_j \|A_j\|_F > \sqrt{d}/2, \qquad \kappa(A) \geq L \exp[(\log d - \log 4)/L] \geq L - \log 4. \tag{1}$$

This is a deterministic energy statement, not a finite-scale rate formula.

Suppose $N \geq 4M_0$. Replicate every teacher row $k = \lfloor N/(2M_0) \rfloor$ times, divide its outgoing coefficient by k, and set all remaining first-layer rows and outgoing coefficients to zero. The teacher computation $g = u \cdot h_1$ is unchanged. Its first energy is at most two, and

$$D = N \|u\|^2 \leq 4M_0 C B^2 = 4T.$$

At least $N/2$ first-layer neurons remain with exact value $q_0 = F(0) = \log 2$, on which u vanishes. These supply a constant without an internal bias.

Write $m = L - 1 \geq 2$ and set

$$q = T^{1/(2m)}, \qquad k_j = \min(j, m - j), \qquad r_j = \frac{q^j}{2\sqrt{T}\, k_j} \quad (1 \leq j \leq m - 1).$$

Then $q^m = \sqrt{T}$, $r_j \le 1/(2qk_j) \le 1/2$, and $r_{m-1} = 1/(2q)$. Choose the positive anchor

$$A = 4 + 3\log(m+2) + \log(T+2), \qquad a_j = \frac{r_j}{r_{j-1}} = q\frac{k_{j-1}}{k_j} \quad (2 \le j \le m-1).$$

The new hidden signal values are

$$h_1^{\text{new}} = F(A + r_1 g), \qquad h_j^{\text{new}} = F(A + a_j(h_{j-1}^{\text{new}} - A)). \tag{2}$$

Implement each new layer using $n = \lfloor N/2 \rfloor$ identical signal neurons and n zero-row constant neurons. Ignore an odd surplus neuron. The coefficients into a signal neuron are a_j on the preceding signal average and $A(1-a_j)/q_0$ on the preceding constant average. Thus every affine expression in (2) is implemented by actual weights and the existing constant channel.

Lemma 2.1. *The final readout $z = 4q(h_{m-1}^{\text{new}} - A)$, implemented using the signal and constant group averages, has signed margin at least one whenever $|g| \ge 1$.*

Proof. At $g = \pm 1$, compare the signal with $A + r_j g$. All additive errors are nonnegative because $F(s) \ge s$, and all corresponding preactivations are at least $A - 1/2$. Each new additive error is therefore at most $e^{-A+1/2}$. Consecutive ideal gains telescope:

$$\prod_{\ell=i+1}^{j} a_\ell = q^{j-i}\frac{k_i}{k_j} \le m\sqrt{T}.$$

The final error is at most

$$m^2\sqrt{T}\, e^{-A+1/2} \le \frac{1}{8q} = \frac{r_{m-1}}{4}.$$

The last inequality follows from $q\sqrt{T} \le T$ and the displayed choice of A. Thus the final output is at least two at $g = 1$ and at most $-3/2$ at $g = -1$. Every map in the scalar chain and the final readout is increasing in g. This extends the margin to all $|g| \ge 1$; an upper bound on the teacher output is not required here. □

Proposition 2.2. *If $m \ge c_0 \log^2(T+2)$, the constructed witness has*

$$\kappa(A_*) \le K := L + C\log^2(LT+2) \tag{3}$$

for a universal constant C.

Proof. The first new signal rows have disjoint teacher and constant supports. Using at least $N/4$ reserved constants, their total energy is at most

$$(D/2)r_1^2 + 2A^2/q_0^2 \le q^2/2 + 2A^2/q_0^2.$$

Each later signal/constant layer has energy exactly $a_j^2 + (A^2/q_0^2)(1-a_j)^2$. The last readout has weighted energy at most $64q^2(1 + A^2/q_0^2)$, including the cost of subtracting A. The factor $N/n \le 3$ handles odd widths.

For $R_j = k_{j-1}/k_j$, the reciprocal pairs $k/(k+1)$ and $(k+1)/k$ give

$$\sum_j R_j^2 \le m + C, \qquad \sum_j (1 - R_j)^2 \le C.$$

Indeed the paired excess of squares above two is $2/[k(k+1)]+k^{-2}+(k+1)^{-2}$, a summable sequence. Since $a_j = qR_j$,

$$\sum_j a_j^2 \le (m+C)q^2, \qquad \sum_j (1-a_j)^2 \le C\{m(q-1)^2+q^2\}.$$

Including the first teacher layer yields

$$\kappa(A_*) \le mq^2 + Cq^2 + CA^2\{1+q^2+m(q-1)^2\}.$$

The depth assumption bounds q and $m(q-1)^2$ by universal constants, while $mq^2 \le m + C\log(T+2)$. Finally $A \le C\log(LT+2)$. This proves (3), with leading coefficient one as in (1). Every constant-channel contribution is charged. □

3 Polynomial operator stability

Lemma 3.1. *There is a polynomial $J = \mathrm{poly}(d, C, B, L)$, independent of $N \ge 4M_0$, such that perturbing every energy-coordinate matrix of the witness by operator norm at most J^{-1} preserves global signed margin $3/4$.*

Proof. Let s, c be the signal and constant group indicator vectors, each of squared norm n. A new internal witness matrix is

$$W_j = sw_j^T, \qquad w_j = (a_j/n)s + [A(1-a_j)/(q_0 n)]c.$$

For every diagonal matrix D_j with entries in $[0,1]$,

$$w_{j+1}^T D_j s = \frac{a_{j+1}}{n} \sum_{i \in \mathrm{signal}} (D_j)_{ii} \in [0, a_{j+1}].$$

Consequently a consecutive product of $D_j W_j$ has operator norm at most

$$\sqrt{a_i^2 + [A(1-a_i)/q_0]^2} \prod_{\ell=i+1}^{j} a_\ell.$$

The telescoping gains bound this by a polynomial in L, T, A. The first injection, original teacher matrix, and final readout have polynomial operator norms, so all relevant propagators are bounded by a polynomial P_0. The bound permits arbitrary intervening diagonal contractions, not merely derivatives at unperturbed states.

The shallow teacher range is polynomial in d, C, B. Also

$$|F(A+y) - A| \le |y| + \log 2 \qquad (A \ge 0).$$

This inequality and the telescoping gains give a polynomial bound Q on every normalized witness state, uniformly on the cube and in N. Write a perturbation block as B_j, with $\|B_j\|_{\mathrm{op}} \le \eta$. The componentwise mean-value formula, using $0 < F' < 1$, gives the exact state-error identity

$$e_j = D_j A_j^* e_{j-1} + D_j B_j (u_{j-1}^* + e_{j-1}),$$

where the D_j are diagonal contractions. Variation of constants bounds the maximum hidden-state error E by

$$E \le \eta P_0 L (Q + E).$$

For $\eta P_0 L \le 1/2$, this gives $E \le 2\eta P_0 L Q$. The final linear readout has polynomial norm, and its own perturbation adds at most $\eta(Q+E)$. Taking a polynomially small η makes total output error at most $1/4$. Lemma 2.1 completes the proof. □

This uses the particular rank-one structure of the planted wire. No polynomial partial-product bound is asserted for arbitrary networks of energy $L + O(\log^2(LT))$.

4 The actual posterior above the width threshold

Fix an input anchor x_0. Center every normalized hidden state of a realized network at that anchor. The resulting scalar activations are

$$a_{b,N}(r) = \frac{F(\sqrt{N}(b+r)) - F(\sqrt{N}b)}{\sqrt{N}}.$$

They fix zero and are 1-Lipschitz. The linear matrices remain A_j, and their product satisfies

$$P(A) \le (\kappa(A)/L)^{L/2}.$$

The centered input has radius at most two. A nonconstant globally margined output has an opposite-sign point, which bounds its anchor offset by $2P(A)$. Constants can be listed separately. Therefore the representation lemma from [1] gives, at energy $K+1$, a common finite sign class with

$$P \le \exp[C\log^2(LT+2)], \qquad \log|\mathcal{F}| \le \exp[C\log^2(d+L+T+2)]. \tag{4}$$

For a finite-scale argument we must justify the posterior normalization, not just compare witness energies. Set $b^2 = 16\log 9 + 8\log(8L)$, and let $\mathcal{T}$ be the intersection of the parameter-block operator balls of radius tb. Since $N \ge 4M_0$ and $M_0 \ge d+1$, a unit-sphere net and Gaussian tails show $\mu(\mathcal{T}) \ge 1/2$. The set is closed, convex, and centrally symmetric. Lemma 3.1 implies that, if $tb \le J^{-1}$, then $A_* + \mathcal{T} \subset E_S$ for every target-labeled sample. Gaussian translation and symmetry yield

$$\mu(E_S) \ge \tfrac{1}{2}e^{-K/(2\sigma^2)}.$$

Gaussian isoperimetry for a set of mass at least $1/2$ then gives

$$Q_S\{\operatorname{dist}(A,\mathcal{T})^2 > K+1\} \le 2e^{-1/(2\sigma^2)}. \tag{5}$$

These statements use the common isotropic variance $\sigma^2 = t^2/N$.

On the complementary event write $A = U + B$ by projection onto $\mathcal{T}$, so that $\kappa(U) \le K+1$ and each B block has operator norm at most $\eta = tb$. Center the *realized* network A, freeze its scalar maps $a_{b,N}$ and its actual anchor output, and replace only the linear matrices by U. The surrogate is allowed to belong to the larger Lipschitz representation class; its anchor maps are not recomputed, and it need not be a native softplus network.

Zero-fixing Lipschitz telescoping and AM–GM bound the output change by

$$2L\eta\left(\frac{(\sqrt{K+1}+\sqrt{L}\eta)^2}{L-1}\right)^{(L-1)/2}. \tag{6}$$

Each of the L terms is bounded by 2η times the other $L-1$ block norms, enlarged by at most η, and their squared sum is at most $(\sqrt{K+1}+\sqrt{L}\eta)^2$. If $\eta \le 1/\sqrt{L(K+1)}$, this sum is at most $K+4$. Using (3), the power in (6) is at most $\exp[C\log^2(LT+2)]$. Thus a condition of the form

$$tb \le \exp[-C\log^2(d+L+T+2)] \tag{7}$$

both preserves the planted witness's margin and makes (6) at most $1/8$.

The surrogate has the same signs as the posterior network on the entire cube, with margin greater than $3/8$. Its product obeys (4). If it is nonconstant, its retained anchor offset is automatically bounded by twice its own product, since its centered computation vanishes at x_0. Hence the same sample-independent class $\mathcal{F}$ contains posterior mass at least $1 - 2e^{-1/(2\sigma^2)}$.

For any fixed $\zeta > 0$, the inequalities $t \le N^{-\zeta/2}$ and $\sigma^2 \le N^{-(1+\zeta)}$ ensure (7) and $2e^{-1/(2\sigma^2)} \le \beta$ above a threshold

$$N_* \le \exp\Big[C_\zeta \log^2(2 + d + C + B + L + \log(2/\beta))\Big], \qquad N_* \ge 4M_0.$$

This is the large-width finite-scale part of the proof. It makes no claim at the critical prefactor $t = 1$.

5 Small widths, cold limits, and learning

For completeness, there is a lower-cost fallback for the purpose of finite-width counting, although it does not match the coefficient in (1). The exact identity

$$F(g) - F(-g) = g$$

provides a complementary relay. At any $N \ge \max(C, 2)$, balance-clone the first teacher rows to width N, dividing outgoing coefficients by their replication counts. The first energy is at most $2d(d+1)B^2$, and the weighted shallow readout energy is at most $2C^2B^2$, independently of N. Split every added relay into two groups of size $\lfloor N/2 \rfloor$ and ignore an odd surplus neuron. Its internal energy is four. The first relay costs at most the weighted shallow readout energy, and the final readout costs at most six. Thus a unit-margin depth-L witness exists with energy polynomial in d, C, B, L, independently of N. For $L = 2$ use the cloned shallow witness directly. If $N = 1$, every represented truth table is constant because the final hidden value is positive; one example determines its sign.

We spell out why the ordinary finite-width energy-net proof also applies to this unbounded activation. Since $F(a) \le \log 2 + |a|$, on an energy ball $\kappa \le R$ the normalized hidden states satisfy

$$\|u_j\| \le \log 2 + \sqrt{R}\,\|u_{j-1}\|\,.$$

Consequently the logarithms of the state bounds and of a uniform parameter-to-output Lipschitz constant Λ_R are $O(L \log(2 + R + L))$, independent of N. Let K_0 be the polynomial fallback witness energy, set $K' = 2K_0 + 2$, and take $r = \min(1, 1/(4\Lambda_{K'}))$. A radius-$r$ witness neighborhood retains signed margin $3/4$. In parameter dimension $p = N(d+2) + (L-2)N^2$, use its inscribed cube with half-side $\min(r/\sqrt{p}, \sigma)$. The cube mass gives

$$\mu(E_S) \ge e^{-K'/(2\sigma^2) - pT_0}, \qquad T_0 = \log\Big(C \max\{1, \sqrt{p/N}/r\}\Big)\,.$$

Clipping at σ removes any divergence in $\log(1/t)$. Dividing the elementary Gaussian energy tail by this denominator puts posterior mass $1 - \beta$ in the ball of radius squared

$$R_\beta = 2K' + \frac{4}{N}\{p(T_0 + \tfrac{1}{2}\log 2) + \log(1/\beta)\}.$$

A net of mesh $1/(8\Lambda_{R_\beta+1})$ there preserves every globally margined sign function. Its log-cardinality is polynomial in $d, N, L, K_0, \log(1/\beta)$. Thus for $N < N_*$ it is quasipolynomial in the resources of

Theorem 1.1. Take the union of these finite classes over the integers $C \le N < N_*$ and the common large-width class. This adds at most $\log(N_* + 1)$ to the largest log-cardinality bound, which remains quasipolynomial. Thus one sample event covers every width and eligible scale at the fixed depth; no width interval is omitted.

For a cold limit at fixed finite architecture, a strict-margin witness of energy K implies

$$Q_S\{\kappa > K + \rho\} \longrightarrow 0 \quad (\sigma \downarrow 0) \qquad \text{for every } \rho > 0.$$

A fixed neighborhood retaining the witness's signs and energy at most $K + \rho/2$ has Gaussian mass of order at least $\sigma^{-p} e^{-(K+\rho/2)/(2\sigma^2)}$, while the tail above $K + \rho$ has the smaller exponential factor $e^{-(K+\rho)/(2\sigma^2)}$ times a polynomial in σ^{-1}. For $N \ge 4M_0$, the matching witness and the class in (4) therefore put every limiting sign posterior on a common class of quasipolynomial log-cardinality. The finite-width argument just given covers the remaining widths uniformly in t and hence in cold limits.

Finally, suppose a sample-independent finite class of size M contains posterior mass at least $1 - \beta$. For $n \ge (\log M + \log(1/\delta))/\alpha$, with probability at least $1 - \delta$ every sample-consistent member has true error at most α. The posterior Gibbs error is then at most $\alpha + \beta$. At a point where correct-label confidence is at most $2/3$, wrong-label probability is at least $1/3$, so $e(S) \le 3(\alpha + \beta)$. Taking $\alpha = \beta = \varepsilon/12$ proves the stated finite-scale bound. For a limiting posterior supported on the class take $\beta = 0$; passing Gibbs error to the limit uses only continuity on the finite truth-table simplex, not continuity of the inclusive confidence threshold. This proves Theorem 1.1.

6 A shallow parity witness and the scope limit

Parity is an example within the shallow teacher class. Put $k(x) = (d + \sum_i x_i)/2 \in \{0, \dots, d\}$. Full parity equals $(-1)^d(-1)^k$, so a fixed readout sign supplies either convention. The alternating values are interpolated by

$$(-1)^k = 1 - 2k + 4\sum_{j=1}^{d-1}(-1)^{j+1}(k - j)_+ \qquad (k \in \{0, \dots, d\}).$$

Replace each hinge by $F(\beta(k - j))/\beta$, whose uniform error is at most $(\log 2)/\beta$, and take $\beta = 32d$. The accumulated error is less than $1/8$. Realize the linear term exactly using $F(k) - F(-k) = k$, and realize the constant using a zero-weight neuron of value $\log 2$. Multiplying the output by two gives signed margin at least one with $d + 2$ neurons and polynomially bounded weights.

When shallow teacher resources are polynomial in learner depth, the depth condition in Theorem 1.1 holds eventually, and its quasipolynomial bound is subexponential in depth. Such shallow softplus targets therefore do not supply exponential-depth lower bounds in the cold or fixed-$\gamma > 1$ regimes considered here. The proof does not cover arbitrary deep teachers, $\gamma = 1$, or arbitrary fixed positive canonical prefactors. In particular, a both-label posterior cannot be normalized by the mass of constant truth tables. These scope limits also prevent any conclusion about the original exact-tanh problem.

References

[1] ChatGPT-6 Astra, K. Hänni, and D. Vaintrob, *Shallow targets under deep logistic posteriors*, companion research proof, 2026. The finite Lipschitz representation lemma and its layer-compression proof are used in Section 1.

[2] C. Borell, *The Brunn–Minkowski inequality in Gauss space*, Inventiones Mathematicae **30** (1975), 207–216.